

NIKKALUOKTA, Sweden

WMO: 020360

Lat: 67.8527N

Lon: 19.0212E

Elev: 1535

StdP: 13.90

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.9	-29.3	-41.0	0.5	-33.6	-35.7	0.8	-29.1	27.5	35.3	24.5	31.9	2.1	310	0.769

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	71.4	57.2	67.8	55.8	64.5	54.4	59.7	67.4	57.8	64.8	56.0	62.3	6.4	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.5	72.1	61.7	54.6	67.1	60.3	52.8	62.8	58.6	26.8	67.7	25.6	64.6	24.4	62.3	66.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.8	18.9	16.0	DB	-39.2	75.9	4.5	3.7	-42.4	78.6	-45.0	80.8	-47.5	82.9	-50.7	85.6	(4)
(5)			WB	-39.2	62.3	4.2	1.7	-42.2	63.6	-44.7	64.6	-47.0	65.6	-50.0	66.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	5.1	5.9	14.4	26.5	38.7	48.4	54.2	50.8	42.4	29.6	14.1	9.5	(6)
(7)		DBStd	20.87	17.06	17.33	13.05	8.90	6.84	6.05	4.86	5.45	6.19	9.79	14.12	15.84	(7)
(8)		HDD50	8168	1393	1235	1104	704	357	101	15	57	237	633	1076	1256	(8)
(9)		HDD65	13348	1858	1655	1569	1154	816	498	335	441	677	1098	1526	1721	(9)
(10)		CDD50	296	0	0	0	0	6	53	146	81	10	0	0	0	(10)
(11)		CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0	(11)
(12)		CDH74	38	0	0	0	0	2	7	24	5	0	0	0	0	(12)
(13)		CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.7	5.1	5.7	6.6	5.8	7.1	6.9	5.5	5.2	5.8	5.7	4.2	5.3	(14)
(15)	Precipitation	PrecAvg	19.6	1.3	1.1	0.9	0.9	1.5	2.4	3.6	2.2	1.9	1.5	1.3	1.2	(15)
(16)		PrecMax	26.9	2.7	2.3	2.2	3.0	3.7	5.6	8.9	5.7	3.9	3.9	3.0	3.4	(16)
(17)		PrecMin	11.1	0.1	0.1	0.2	0.0	0.2	0.5	0.9	0.4	0.6	0.0	0.0	0.4	(17)
(18)		PrecStd	4.1	0.7	0.6	0.5	0.6	0.9	1.1	1.7	1.3	1.0	0.9	0.7	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.3	42.1	43.3	52.5	70.1	73.6	77.4	73.7	63.3	51.8	45.2	43.4	(19)
MCWB			36.7	36.6	36.6	43.0	51.4	56.2	59.8	59.4	53.3	46.3	40.1	37.8	(20)	
2%		DB	39.1	38.7	39.6	45.5	61.5	68.5	72.2	69.6	58.6	48.7	40.9	39.4	(21)	
		MCWB	34.7	34.1	34.6	38.7	48.4	53.3	58.4	57.8	51.2	43.9	36.8	34.9	(22)	
5%		DB	34.4	35.6	36.6	42.4	54.7	64.6	68.1	65.8	55.9	46.4	36.9	34.9	(23)	
		MCWB	30.9	31.9	32.3	36.5	45.3	52.2	57.0	56.8	49.7	41.9	33.8	31.2	(24)	
10%		DB	29.8	30.9	33.2	39.9	50.3	60.7	64.7	62.0	53.4	43.2	33.2	31.4	(25)	
		MCWB	27.0	27.9	29.5	34.7	42.6	50.7	55.1	54.5	48.1	39.2	31.0	28.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.2	37.6	37.1	43.1	52.8	58.9	62.6	62.1	54.6	46.8	40.2	38.3	(27)
MCD B			41.7	41.9	42.1	52.7	67.3	69.2	72.0	70.3	60.0	50.5	44.7	42.8	(28)	
2%		WB	34.7	34.5	34.9	39.2	49.1	55.7	60.3	59.4	52.5	44.2	37.3	35.1	(29)	
		MCD B	38.9	38.4	39.3	44.6	60.0	65.1	68.1	66.8	57.3	48.2	40.8	39.3	(30)	
5%		WB	31.1	31.9	32.6	37.1	45.9	53.5	58.5	57.4	50.5	42.0	33.9	31.4	(31)	
		MCD B	34.3	35.8	36.5	41.7	54.6	61.8	65.6	64.4	54.8	46.0	36.4	34.6	(32)	
10%		WB	27.0	28.3	29.6	35.1	43.1	51.3	56.8	55.4	48.7	39.3	31.2	28.6	(33)	
		MCD B	29.8	31.1	33.1	39.3	49.5	59.2	62.9	61.2	52.7	42.8	33.2	31.4	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.8	20.1	21.8	18.4	14.5	15.1	15.7	16.9	15.1	12.8	15.7	18.3	(35)
MCD BR			17.9	19.0	17.5	19.2	24.7	23.7	23.9	25.2	20.5	14.2	16.9	17.6	(36)	
MCWBR			15.4	15.7	14.3	13.5	14.0	11.6	11.8	14.5	13.6	10.6	13.9	15.4	(37)	
5% WB		MCD BR	17.7	18.8	16.6	17.0	23.3	20.6	20.1	22.4	17.0	12.7	15.4	17.0	(38)	
		MCWBR	15.6	16.3	13.9	12.1	13.5	10.9	10.9	13.4	11.8	9.8	12.9	15.0	(39)	
																(39)
(40)	Clear-Sky Solar Irradiance	taub	0.173	0.233	0.275	0.302	0.320	0.324	0.332	0.319	0.287	0.251	0.177	N/A	(40)	
taud		1.921	2.117	2.187	2.228	2.363	2.466	2.493	2.510	2.563	2.284	1.962	N/A	(41)		
Ebn at Noon		44	186	243	269	277	279	273	265	248	180	38	0	(42)		
Edh at Noon		3	15	25	32	31	30	28	24	18	12	3	0	(43)		
(44)	All-Sky Solar Radiation	RadAvg	12	148	599	1220	1571	1576	1452	1060	563	188	32	0	(44)	
(45)		RadStd	1	16	35	58	121	130	145	99	59	27	3	0	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page